

Business Requirements Document: Research AI Platform

Project Name: Research AI Platform

Application Type: AI-powered research assistant and scientific knowledge workspace

Target Users: Researchers, students, academic writers, R&D teams, analysts, and knowledge workers

Platform URL: <http://localhost:5173/dashboard>

Executive Summary

The Research AI Platform is a full stack AI-powered dashboard that helps users upload research documents, search through indexed knowledge, ask grounded questions, generate summaries, manage references, organize scholars, and maintain research chat history. The platform reduces manual research effort by combining document processing, semantic search, RAG-based chat, citation support, and structured research workflows in one authenticated workspace.

The system is designed as a modern production-ready application using a React frontend, FastAPI backend, authenticated APIs, database-backed persistence, document indexing, AI response generation, and multiple chat modes including Lite Mode, Deep Review, and Source Mode.

Business Problem

Researchers spend significant time reading papers, extracting key findings, comparing methods, managing citations, and organizing literature. Traditional workflows are fragmented across PDFs, browser tabs, citation tools, notes, and search engines. This causes:

- Slow literature review preparation
- Difficulty finding relevant content inside uploaded PDFs
- Poor traceability between AI answers and document sources
- Repeated manual summarization
- Lack of centralized research memory
- Inefficient organization of authors, papers, topics, and reading lists

Business Objective

The objective is to deliver a centralized AI research workspace that allows users to move from document upload to search, chat, citation, and research organization without leaving the dashboard.

The platform must help users:

- Upload and index research documents

- Ask AI questions over uploaded content
- Retrieve source-grounded answers
- Search documents semantically
- Save chat history per user
- Build research plans and reading workflows
- Manage papers, authors, citations, and knowledge entries
- Use Lite Mode for fast answers and advanced modes for deeper research

Core Modules

1. Authentication

The platform must require users to register and log in before accessing the dashboard. Each user's chats, library items, scholar entries, uploads, and search results must be isolated by account.

Key requirements:

- User registration
- Login with secure password handling
- JWT-based authenticated sessions
- Protected dashboard routes
- Logout support
- User-specific data filtering on backend APIs

2. Dashboard

The dashboard must act as the main research workspace. It should provide a sidebar-based navigation system with access to major research modules.

Required dashboard sections:

- New Chat
- Search
- Chat History
- Science Navigator
- Library
- Scholars

- SciencePedia
- Knowledge Base
- Practice Lab

The dashboard should feel like a professional scientific workspace, not a simple chatbot page.

3. New Chat

The system must provide a visible **New Chat** option from the dashboard sidebar. When clicked, it should reset the current conversation and open a fresh research chat workspace.

Business requirements:

- Start a new conversation without logging out
- Clear current chat state
- Preserve previous saved conversations in history
- Default new chats to Lite Mode
- Allow users to change mode before sending a message
- Keep UI responsive during chat reset

4. Research Chat

The Research Chat module is the core AI interface of the platform. Users should be able to ask questions, upload files, retrieve sources, generate summaries, and export conversations.

Required features:

- AI question answering
- File upload from chat
- Drag-and-drop upload
- Multi-message conversation UI
- Loading state while generating answers
- Copy assistant responses
- Regenerate answers
- Helpful / not helpful feedback
- Export chat as Markdown
- Source reference panel

- Citation style selection
- Top-K retrieval control
- Browser and backend cache support

5. Chat Modes

The system must support multiple chat modes.

Lite Mode

Lite Mode is the default fast-response mode for quick research questions. It should provide concise answers using limited context and fewer retrieved chunks.

Expected behavior:

- Fastest answer mode
- Concise response in 1 to 3 short paragraphs
- Lower context usage
- Lower compute cost
- Best for quick summaries, definitions, and direct questions
- Should still use uploaded document context when available

Deep Review Mode

Deep Review Mode should generate structured research analysis.

Expected behavior:

- Detailed synthesis
- Key findings
- Evidence from sources
- Limitations
- Practical next steps
- Better for literature review work

Source Mode

Source Mode should prioritize strict traceability.

Expected behavior:

- Citation-first answers

- Source-numbered bullets
- Minimal unsupported claims
- Useful for academic writing and evidence extraction

6. Lite Mode Limitations / Not Fully Working Items

In the current product scope, Lite Mode should not be expected to perform heavy research workflows. These should either be disabled, shown as limited, or upgraded to Deep Review / Source Mode.

Lite Mode limitations:

- Does not perform full literature review synthesis
- Does not deeply compare many papers
- Does not guarantee exhaustive citation coverage
- Does not use large context windows
- Does not perform external academic web search unless connectors are configured
- Does not generate full research reports
- Does not build knowledge graphs
- Does not perform long-running multi-agent research
- Does not guarantee page-level citation precision for every claim
- Does not replace Source Mode for strict evidence tracing

Recommended product behavior:

- Show Lite Mode as “Fast answer with key references”
- Display a warning when users request deep review tasks in Lite Mode
- Suggest switching to Deep Review or Source Mode when needed
- Backend must enforce Lite Mode limits, not only frontend UI

7. Retrieval Scope

The chat module must support retrieval scope selection.

Scopes:

- Document: uploaded and indexed files only
- Web: external academic providers when configured
- Hybrid: documents first, web sources when configured

Current limitation:

- Web and Hybrid modes may fall back to indexed documents if academic connectors such as CrossRef, Semantic Scholar, PubMed, or arXiv are not configured.

8. Semantic Search

The Search module must allow users to search through their uploaded and indexed research documents.

Requirements:

- Search only the signed-in user's indexed content
- Return relevant snippets
- Show filename and page/section metadata when available
- Display relevance score
- Support quick prompt buttons
- Show empty state when no matches exist
- Handle backend errors gracefully

9. Document Upload and Indexing

The platform must allow users to upload research files for AI processing.

Supported file types:

- PDF
- DOCX
- TXT
- PPTX
- CSV

Requirements:

- Upload files from chat or library
- Extract text content
- Chunk documents
- Store indexed content for retrieval
- Generate summary after upload
- Clear stale cache after new upload

- Show upload progress
- Return file name, file type, page count or section count, chunk count, and summary

10. Library

The Library module must allow users to manage research materials.

Required features:

- Upload file
- Add paper manually
- Add URL or DOI
- Import BibTeX or RIS
- Search paper metadata placeholder
- Prepare Zotero import workflow
- View saved library items
- Generate BibTeX export for saved items

Business value:

- Helps users create a reusable research collection
- Supports citation workflows
- Connects uploaded documents with future search and chat use

11. Scholars Workspace

The Scholars module must help users organize research entities and reading workflows.

Required features:

- Track important authors
- Save author focus areas
- Group papers by theme, method, or dataset
- Build topic-specific reading lists
- Store scholar entries per user

Business value:

- Supports literature review planning
- Helps users organize research around authors, themes, and papers

- Adds recruiter-friendly product depth beyond basic AI chat

12. Science Navigator

Science Navigator must help users turn a broad topic into a structured research plan.

Required output:

- Research questions
- Methods to investigate
- Dataset suggestions
- Next research directions

Business value:

- Helps users start research faster
- Converts vague topics into actionable plans
- Supports academic and R&D workflows

13. Chat History

The platform must save successful chat interactions per user.

Requirements:

- Store message, response, mode, and user ID
- Display conversations in reverse chronological order
- Show chat mode label
- Allow search in history
- Allow delete by owner only
- Do not expose one user's history to another user

14. Citation and References

The platform must support citation-oriented research workflows.

Requirements:

- Show references in a side panel
- Display source title, page, confidence, citation, and quote when available
- Support citation styles:
 - Numbered

- Inline
- APA
- IEEE
- Include source chips inside assistant answers
- Export chat with sources in Markdown

15. Caching

The platform must support multi-level caching to improve response time and reduce AI cost.

Required cache layers:

- Browser local cache
- Redis or backend JSON cache
- Database cache from previous chat answers

Expected behavior:

- Cache can be turned off for fresh generation
- Cache should consider message, user ID, mode, scope, citation style, and Top-K
- Cache should be cleared when new documents are uploaded
- UI should show cache layer when applicable

16. Non-Functional Requirements

Performance:

- Lite Mode should return faster than Deep Review and Source Mode
- Search should return results quickly for indexed documents
- UI must remain responsive during upload and answer generation

Security:

- All dashboard APIs must require authentication
- User data must be isolated by user ID
- Passwords must be securely hashed
- JWT tokens must be handled safely
- Backend must validate inputs and ownership

Scalability:

- Backend should support modular services for chat, upload, search, library, scholars, and cache
- Retrieval and generation logic should remain separate from API routing
- Database schema should support future project/team workspaces

Reliability:

- Failed AI generation should not save bad history
- Upload errors must be clearly shown
- Empty states must guide users without breaking workflows
- API errors must return user-friendly messages

Usability:

- Dashboard navigation must be clear
- New Chat must be obvious
- Lite Mode must be visually identifiable
- Disabled or unavailable features must explain why
- Users should never wonder whether they are searching documents, web, or hybrid sources

Production Acceptance Criteria

The project is production-ready when:

- A user can register, log in, and access the dashboard
- A user can start a new chat in Lite Mode
- A user can upload a document and receive an indexed summary
- A user can ask questions over uploaded documents
- Chat responses are saved to history by user
- Search returns user-specific document results
- Library items can be created and exported as BibTeX
- Scholar entries can be saved and retrieved
- Lite Mode, Deep Review, and Source Mode behave differently
- Web/hybrid limitations are clearly communicated
- Unauthorized users cannot access protected APIs

- Backend validates message, mode, scope, and user ownership
- UI handles loading, empty, error, and success states
- Build passes without production errors

Recruiter-Focused Project Positioning

This project demonstrates strong full stack engineering skills because it includes:

- React dashboard architecture
- Component-based UI design
- Protected routing
- JWT authentication
- FastAPI backend development
- SQLAlchemy database models
- User-specific data isolation
- AI/RAG workflow design
- Document upload and indexing
- Semantic search integration
- Chat history persistence
- Citation-aware AI responses
- Multi-mode AI behavior
- Caching strategy
- Production-style error handling
- Research-focused product thinking

Professional Summary

The Research AI Platform is a production-oriented AI research workspace that combines document intelligence, semantic search, citation-grounded chat, and research organization tools. It goes beyond a basic chatbot by supporting authenticated users, persistent history, document upload, library management, scholar workflows, multiple AI modes, and scalable backend APIs. Lite Mode provides a fast and cost-efficient entry point, while Deep Review and Source Mode support advanced academic research workflows.